

A Scalar Counterexample to the Positive Lower-Ratio Approximation Conjecture

Automatic math-discovery run `fa_banach_001`

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Status

Candidate full counterexample to the lower-ratio part of Remark 2 in Chang’s arXiv:2407.05062v1 [1]. The proof is exact and one-dimensional. Human review is recommended, chiefly to decide whether the source intended an unstated strict-positivity hypothesis on the numerator.

Source problem

Section 7.1 of the source permits an arbitrary real-valued continuous function f and a polynomial map Φ of the broad form in its Eq. (10). It asks for an invertible denominator g and polynomial arguments satisfying the single-operator upper and lower ratios (74)–(75), and the averaged versions (81)–(82). Remark 2 then says that the authors “conjecture the existence of function $g(\mathbf{x})$ and polynomials in terms of Ψ as arguments” for any prescribed α_1 or α_2 [1, p. 29]. Both α_1 and α_2 are specified to be positive.

The relevant setup and four requirements are reproduced from PDF pages 23–25, followed by Remark 2 on page 29.

one input variable to multiple variables.

7 Multivariate Hypercomplex Function Approximation

In this section, we will consider multivariate hypercomplex function approximation problem in terms of the ratio error discussed in Section 7.1 and the difference error in Section 7.2

7.1 Ratio Type Approximation

Let $\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_n$ be n self-adjoint operators with $\Lambda(\mathbf{A}_1) \times \Lambda(\mathbf{A}_2) \times \dots \times \Lambda(\mathbf{A}_n) \in \bigtimes_{i=1}^n [m_i, M_i] \in \mathbb{R}^n$ for real scalars $m_i < M_i$. The mapping $\Phi : \mathbb{B}(\mathcal{H}) \rightarrow \mathbb{B}(\mathcal{K})$ is defined by Eq. (10). Let $f(\mathbf{x})$ be any real valued continuous functions with n input variables defined on the range $\bigtimes_{i=1}^n [m_i, M_i]$, represented by $f(\mathbf{x}) \in \mathcal{C}\left(\bigtimes_{i=1}^n [m_i, M_i]\right)$. The *ratio type approximation* problem is to find the invertible function $g(\mathbf{x})$ and n polynomial type functions, represented by $\text{Poly}_{f,\Psi,\mathcal{U}}(\underline{\mathbf{A}}_i)$ for $i = 1, 2, \dots, n$, to satisfy the following:

$$\frac{\Phi(f(\underline{\mathbf{A}}))}{g(\mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{U}}(\underline{\mathbf{A}}_1) \mathbf{V}, \dots, \mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{U}}(\underline{\mathbf{A}}_n) \mathbf{V})} \leq \alpha_1 \mathbf{I}_{\mathcal{K}}, \quad (74)$$

where α_1 is some specified positive real number. Similarly, we also can find the invertible function $g(\mathbf{x})$ and n polynomial type functions, represented by $\text{Poly}_{f,\Psi,\mathcal{L}}(\underline{\mathbf{A}}_i)$ for $i = 1, 2, \dots, n$, to satisfy the following:

$$\frac{\Phi(f(\underline{\mathbf{A}}))}{g(\mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{L}}(\underline{\mathbf{A}}_1) \mathbf{V}, \dots, \mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{L}}(\underline{\mathbf{A}}_n) \mathbf{V})} \geq \alpha_2 \mathbf{I}_{\mathcal{K}}, \quad (75)$$

where α_2 is some specified positive real number.

From Theorem 4, if $g(\mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{U}}(\underline{\mathbf{A}}_1) \mathbf{V}, \dots, \mathbf{V}^* \text{Poly}_{f,\Psi,\mathcal{U}}(\underline{\mathbf{A}}_n) \mathbf{V}) > \mathbf{0}$, Eq. (74) can be established by setting

$$\alpha_1 \geq \left[\max_{x \in \left(\widetilde{\text{Poly}}_{f,\Psi,\mathcal{U}}(\mathbf{I}^\times) \right), x \in \bigtimes_{i=1}^n \left(\widetilde{\text{Poly}}_{f,\Psi,\mathcal{U}}(\mathbf{I}_i^\times) \right)} x g^{-1}(x) \right] \quad (76)$$

Source page 23: the single-operator requirements (74)–(75).

$$\alpha_2 \leq \left[\max_{x \in \left(\widetilde{\text{Poly}}_{f, \Psi, \mathcal{U}}(\mathbf{I}^\times) \right)} \prod_{i=1}^n \left(\widetilde{\text{Poly}}_{f, \Psi, \mathcal{U}}(\mathbf{I}_i^\times) \right) x g^{-1}(\mathbf{x}) \right]. \quad (79)$$

Let $\mathbf{A}_{j_i}$ be self-adjoint operators with $\Lambda(\mathbf{A}_{j_i}) \in [m_i, M_i]$ for real scalars $m_i < M_i$. The mappings $\Phi_{j_1, \dots, j_n} : \mathcal{B}(\mathfrak{H}) \rightarrow \mathcal{B}(\mathfrak{K})$ are defined by Eq. (10), where $j_i = 1, 2, \dots, k_i$ for $i = 1, 2, \dots, n$. We have n probability vectors $\mathbf{w}_i = [w_{i,1}, w_{i,2}, \dots, w_{i,k_i}]$ with the dimension k_i for $i = 1, 2, \dots, n$, i.e., $\sum_{\ell=1}^{k_i} w_{i,\ell} = 1$.

Let $f(\mathbf{x})$ be any real valued continuous functions with n variables defined on the range $\prod_{i=1}^n [m_i, M_i] \in \mathbb{R}^n$, where $\times$ is the Cartesian product. Besides, given any $\epsilon > 0$, we assume that the function $f(\mathbf{x})$ satisfies the following:

$$\begin{aligned} 0 &\leq \Psi_{\mathcal{U}}(\mathbf{x}) - f(\mathbf{x}) \leq \epsilon, \\ 0 &\leq f(\mathbf{x}) - \Psi_{\mathcal{L}}(\mathbf{x}) \leq \epsilon, \end{aligned} \quad (80)$$

for $\mathbf{x} \in \mathbf{I}^\times$. The function $g(\mathbf{x})$ is also a real valued continuous function with n variables defined on the range $\prod_{i=1}^n [m_i, M_i]$. The *ratio type approximation* problem is to find the invertible function $g(\mathbf{x})$ and n

arguments of the function $g(\mathbf{x})$ with the format $\sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{U}}(\mathbf{A}_{j_i}) \mathbf{V}$ for $i = 1, 2, \dots, n$, to satisfy the following:

$$\frac{\sum_{j=1}^k \mathbf{w}_j^\times \Phi_j(f(\mathbf{A}_j))}{g\left(\sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{U}}(\mathbf{A}_{j_1}) \mathbf{V}, \dots, \sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{U}}(\mathbf{A}_{j_n}) \mathbf{V}\right)} \leq \alpha_1 \mathbf{I}_{\mathfrak{K}}, \quad (81)$$

where α_1 is some specified positive real number. Similarly, we also can find the invertible function $g(\mathbf{x})$ and n arguments of the function $g(\mathbf{x})$ with the format $\sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{L}}(\mathbf{A}_{j_i}) \mathbf{V}$ for $i = 1, 2, \dots, n$, to

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Source page 24: the averaged setup and requirement (81).

satisfy the following:

$$\frac{\sum_{j=1}^k \mathbf{w}_j^\times \Phi_j(f(\mathbf{A}_j))}{g\left(\sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{L}}(\mathbf{A}_{j_1}) \mathbf{V}, \dots, \sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{L}}(\mathbf{A}_{j_n}) \mathbf{V}\right)} \geq \alpha_2 \mathbf{I}_{\mathfrak{K}}, \quad (82)$$

where α_2 is some specified positive real number.

From Theorem 4, if $g\left(\sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{U}}(\mathbf{A}_{j_1}) \mathbf{V}, \dots, \sum_{j=1}^k \mathbf{w}_j^\times \mathbf{V}^* \text{Poly}_{f, \Psi_j, \mathcal{U}}(\mathbf{A}_{j_n}) \mathbf{V}\right) > 0$, Eq. (81) can be established by setting

Source page 25: the averaged lower requirement (82).

$$\alpha_2 \leq \left[\min_{\substack{m_i \leq x_i \leq M_i \\ i=1,2,\dots,n}} \frac{\sum_{i=1}^n a_i x_i + b}{\exp\left(\sum_{i=1}^n \beta_i x_i\right)} \right]. \quad (99)$$

Remark 2 Given requirements of Eq. (74), Eq. (75), Eq. (81), and Eq. (82), we conjecture the existence of function $g(\mathbf{x})$ and polynomials in terms of Ψ as arguments: $\text{Poly}_{f,\Psi,\mathfrak{U}}(\mathbf{x})$, $\text{Poly}_{f,\Psi_j,\mathfrak{U}}(\mathbf{x})$ and $\text{Poly}_{f,\Psi,\mathcal{L}}(\mathbf{x})$, $\text{Poly}_{f,\Psi_j,\mathcal{L}}(\mathbf{x})$ for any given α_1 or α_2 . If existence, how to find these functions?

Source page 29: Remark 2.

Precise claim addressed

The lower single-operator requirement is, in the scalar specialization,

$$\frac{\Phi(f(A))}{g(P_L(A))} \geq \alpha_2, \quad \alpha_2 > 0, \quad (1)$$

where $P_L(A)$ abbreviates the proposed lower polynomial argument and the denominator is required to be invertible. The averaged requirement replaces the numerator and the argument by the weighted expressions in Eq. (82). Remark 2 asserts existence for arbitrary prescribed α_2 in this setup. Refuting either lower requirement refutes that universal assertion.

Idea of the proof

The polynomial maps admitted by the source can annihilate a strictly positive input. Choose $f \equiv 1$ but take $\Phi(X) = X^2 - X^3$, so that $\Phi(f(A)) = \Phi(1) = 0$. Changing g or any of its polynomial arguments can only change the denominator. An invertible denominator cannot turn a zero numerator into a strictly positive ratio. Taking a one-term average makes the averaged formula identical to the same scalar contradiction.

Counterexample

Theorem 1. *For every $\alpha_2 > 0$, there are data satisfying the stated setup of Section 7.1 for which no invertible g and no choice of polynomial arguments can satisfy Eq. (75). The same data, with a one-term average, also make Eq. (82) impossible. Consequently the positive lower-ratio existence part of Remark 2 is false as stated.*

Proof. Take

$$n = 1, \quad \mathfrak{H} = \mathfrak{K} = \mathbb{C}, \quad I = [0, 1], \quad A = \frac{1}{2}, \quad V = 1,$$

and let $f(x) = 1$ on I . Define the polynomial map

$$\Phi(X) = X^2 - X^3. \quad (2)$$

This is exactly of the source's Eq. (10) form: take $a_2 = 1$, $a_3 = -1$, all other coefficients zero, and $V = 1$. The source expressly imposes no linearity, positivity, or normalization constraint on this generalized Φ [1, p. 5].

For completeness, the sigmoid bracketing assumptions used in the surrounding results are nonempty for these data. Given $\varepsilon > 0$, let $\sigma(t) = (1 + e^{-t})^{-1}$ and choose $0 < \delta < \min\{1, \varepsilon\}$. Then

$$\Psi_U(x) = 1 + \delta\sigma(x), \quad \Psi_L(x) = 1 - \delta\sigma(x)$$

are linear combinations of sigmoids (use $1 = 2\sigma(0)$), remain strictly positive on I , and satisfy

$$0 \leq \Psi_U - f \leq \varepsilon, \quad 0 \leq f - \Psi_L \leq \varepsilon.$$

Functional calculus gives $f(A) = 1$. Applying (2),

$$\Phi(f(A)) = \Phi(1) = 1^2 - 1^3 = 0. \tag{3}$$

Now allow *any* proposed lower polynomial argument $P_L(A)$ and *any* proposed g for which the scalar denominator $d = g(P_L(A))$ is invertible. Thus $d \neq 0$, but (3) yields

$$\frac{\Phi(f(A))}{g(P_L(A))} = \frac{0}{d} = 0 < \alpha_2.$$

This contradicts (1) independently of how g and P_L are chosen.

For Eq. (82), take $k_1 = 1$, weight $w_{1,1} = 1$, and the same A , Φ , f , and brackets. Its weighted numerator is again $\Phi(f(A)) = 0$, so the identical contradiction proves the averaged assertion. $\square$

Structural obstruction

The example is not an artifact of choosing an unfortunate scale for g . It realizes the exact missing hypothesis.

Proposition 1. *Under the positive-denominator interpretation used in the source theorem, if*

$$G^{-1/2}YG^{-1/2} \geq \alpha I, \quad G > 0, \quad \alpha > 0,$$

then $Y \geq \alpha G > 0$. In particular, a singular numerator Y can never satisfy a positive lower-ratio bound.

Proof. Congruence by $G^{1/2}$ gives $Y \geq \alpha G$. Since G is positive and invertible, its lower spectral bound is strictly positive, and hence so is that of Y . $\square$

The scalar construction in Theorem 1 avoids any possible ambiguity about noncommutative quotient notation. It shows that a repair must at least restrict the allowed data so that $\Phi(f(A))$ has a uniform positive lower bound. That condition is absent from Remark 2.

Verification, scope, and novelty

The accompanying script checks the scalar identity with exact rational arithmetic for representative positive and negative nonzero denominators. It is a sanity check only; the proof does not depend on computation.

This packet refutes the lower-ratio requirements (75) and (82), which is enough to disprove Remark 2 as a universal existence claim for arbitrary positive α_2 . It does not claim that the upper-ratio requirement for α_1 is false. It also does not resolve the separate difference-type Remark 3; its displayed requirements contain a free scalar x on the right-hand side and need clarification before a precise universal statement can be assessed.

A bounded novelty search on August 11, 2026 covered the run indexes, the exact arXiv id and title, the phrases “ratio type approximation” and the conjecture wording, and current web/arXiv results for the author and topic. It found the source and mirrors but no later paper explicitly resolving Remark 2. Absence from that bounded search is not proof of novelty.

Human-review recommendation

Send to human review. The decisive checks are that Eq. (10) really admits (2), and that the printed conjecture contains no strict-positivity hypothesis on $\Phi(f(A))$. If such a hypothesis was intended, it should be stated as a repair; it is not part of the conjecture addressed here.

References

- [1] S.-Y. Chang, *Generalized Multivariate Hypercomplex Function Inequalities and Their Applications*, arXiv:2407.05062v1 (2024), <https://arxiv.org/abs/2407.05062>.